

LECTURE-10

MATHEMATICS 1 (15B11MA111)

Maxima and Minima of Functions of Two Variables

Partial Differentiation CO2 [C105.2]

Content:

Maxima and Minima of Functions of Two Variables

- Introduction
- Necessary Condition for Maxima and Minima
- Sufficient Condition for Maxima and Minima
- Numerical Problems

2 Reference

- R.K. Jain and S.R.K. Iyenger, "Advanced Engineering Mathematics" fifth edition, Narosa publishing house, 2016.

Introduction

Minima/Minimum of a Function:

Let $f(x, y)$ be a function defined in a closed and bounded region R . Let (a, b) be an interior point of R and $(a + h, b + k)$ be a neighborhood of (a, b) lying inside R .

Then (a, b) is called a local (relative) minima of $f(x, y)$ if

$$f(a, b) \leq f(a + h, b + k), \quad \forall h, k.$$

or

$$f(a + h, b + k) - f(a, b) \geq 0.$$

and the value $f(a, b)$ is called local (relative) minima or the minimum value of f in the neighborhood of (a, b) .

Cont...

Maxima/Maximum of a Function

Let $f(x,y)$ be a function defined in a closed and bounded region R . Let (a,b) be an interior point of R and $(a+h,b+k)$ be the neighborhood of (a,b) lying inside R .

Then (a,b) is called a local (relative) maxima of $f(x,y)$ if

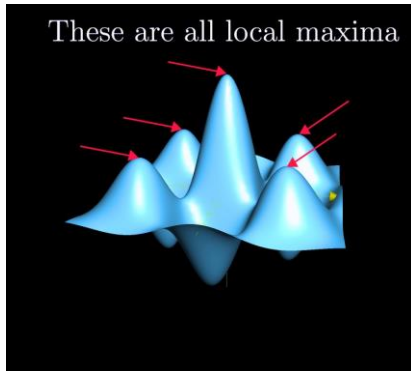
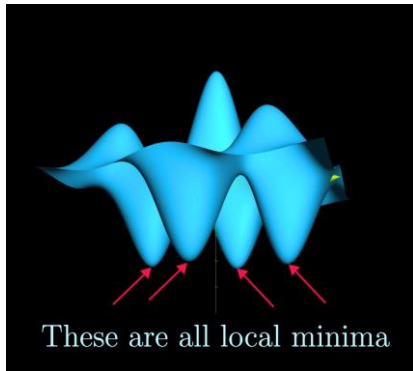
$$f(a,b) \geq f(a+h,b+k), \quad \forall h, k.$$

or

$$f(a+h,b+k) - f(a,b) \leq 0.$$

The value $f(a,b)$ is called local (relative) maxima or the maximum value of f in the neighborhood of (a,b) .

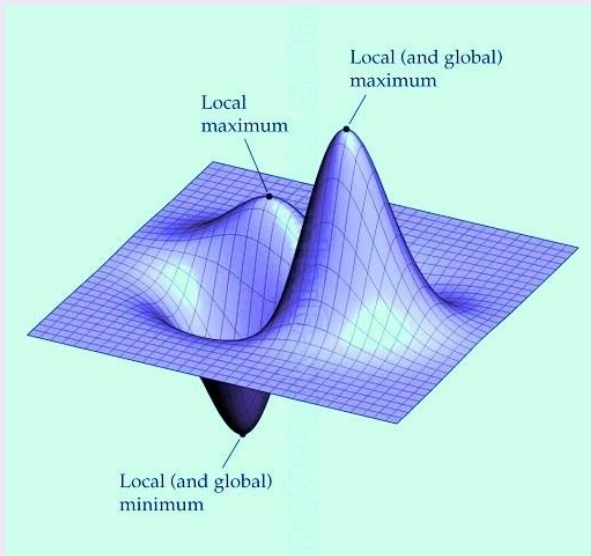
Local Minima and Maxima:



Global Minima and Maxima

- ❑ The smallest and largest value attained by the function f over the entire region R including the boundary are called global (absolute) minima and global (absolute) maxima respectively.
- ❑ The points at which the relative minimum and relative maximum values of the function are attained are called **extreme points** or **stationary points** or **critical point** and corresponding minimum and maximum values are called extreme values of the function.

Global Minima and Maxima



Necessary Condition for Extreme Points

Let the function $f(x, y)$ be continuous and possess first order partial derivatives at a point $P(a, b)$. Then the necessary condition for the existence of extreme values at this point is

$$f_x(a, b) = 0, \quad f_y(a, b) = 0.$$

Proof: Let $(a + h, b + k)$ be the neighborhood of (a, b) , then Taylor's series expansion of $f(x, y)$ about the point (a, b) is given as

$$\begin{aligned} f(a + h, b + k) &= f(a, b) + hf_x(a, b) + kf_y(a, b) \\ &\quad + \frac{1}{2} \left(h^2 f_{xx}(a, b) + 2hkf_{xy}(a, b) + k^2 f_{yy}(a, b) \right) + \dots \end{aligned}$$

Neglecting the second and higher degree terms of h and k , we get

$$\Delta f = f(a + h, b + k) - f(a, b) = hf_x(a, b) + kf_y(a, b)$$

Cont...

The sign of $\Delta f (\leq 0 \text{ or } \geq 0)$ depends on the expression

$$hf_x(a,b) + kf_y(a,b)$$

which is a function of h and k . Letting $h \rightarrow 0$, we find that Δf changes sign with that of k . Therefore

the function can not have extrema unless $f_y = 0$.

Similarly letting $k \rightarrow 0$, we find that Δf changes sign with that of h . Therefore

the function can not have extrema unless $f_x = 0$.

Therefore the necessary condition for a function to have extrema at a point $P(a,b)$ is that $f_x(a,b) = 0$ and $f_y(a,b) = 0$.

A point (a,b) where $f_x = 0$ and $f_y = 0$ is called stationary/critical point.

Sufficient Condition for Extreme Points

Let the function $f(x, y)$ be continuous and possess first and second order partial derivatives at a point $P(a, b)$. Let $P(a, b)$ be a critical point, then the point $P(a, b)$ is a point of

- relative minima if $rt - s^2 > 0$ and $r > 0$,
- relative maxima if $rt - s^2 > 0$ and $r < 0$.

where

$$r = f_{xx}(a, b), s = f_{xy}(a, b), t = f_{yy}(a, b).$$

Note: No conclusion can be drawn from the case $rt - s^2 = 0$, further investigation is needed in this case.

If $rt - s^2 < 0$, then the function has neither minimum nor maximum at the point $P(a, b)$ and this point is called the saddle point.

Cont...

Proof: Let $(a+h, b+k)$ be neighborhood of the point $P(a, b)$, then Taylor's series expansion of $f(x, y)$ about the point (a, b) is given as

$$\begin{aligned} f(a+h, b+k) &= f(a, b) + hf_x(a, b) + kf_y(a, b) \\ &\quad + \frac{1}{2} \left(h^2 f_{xx}(a, b) + 2hkf_{xy}(a, b) + k^2 f_{yy}(a, b) \right) + \dots \end{aligned}$$

Since $P(a, b)$ is a critical point, therefore we have $f_x(a, b) = 0$ and $f_y(a, b) = 0$ and neglecting the third and the higher order terms of h and k , we get

$$\begin{aligned} \Delta f &= f(a+h, b+k) - f(a, b) = \frac{1}{2} (h^2 r + 2hks + k^2 t) \\ &= \frac{1}{2r} (h^2 r^2 + 2hk sr + k^2 tr + k^2 s^2 - k^2 s^2) \\ &= \frac{1}{2r} ((hr + ks)^2 + k^2 (rt - s^2)) \end{aligned}$$

Cont...

Since at the point $P(a,b)$ the function $f(x,y)$ has a local minima if

$$\Delta f = f(a+h, b+k) - f(a,b) \geq 0$$

and local maxima if

$$\Delta f = f(a+h, b+k) - f(a,b) \leq 0$$

Also, the sufficient condition for $((hr + ks)^2 + k^2(rt - s^2))$ to be positive is that

$$rt - s^2 > 0.$$

Now sign of Δf depends on the sign of r .

Cont...

Since $\Delta f > 0$ if $r > 0$, therefore f has a local minima at (a, b) if

$$rt - s^2 > 0, \quad r > 0.$$

and $\Delta f < 0$ if $r < 0$, therefore f has a local maxima at (a, b) if

$$rt - s^2 > 0, \quad r < 0.$$

If $rt - s^2 < 0$, then sign of Δf depends on the sign of h and k , therefore neither minima nor maxima is attained on $P(a, b)$.

If $rt - s^2 = 0$, then we find that $r = s = t = 0$, therefore no conclusion can be drawn, so terms involving higher order partial derivatives must be considered.

Example: Find the relative minimum and maximum value of the function

$$f(x,y) = 2(x^2 - y^2) - x^4 + y^4.$$

Solution: We have

$$f_x = 4x - 4x^3 = 0 \Rightarrow 4x(1 - x^2) = 0 \Rightarrow x = 0, \pm 1.$$

$$f_y = -4y + 4y^3 = 0 \Rightarrow -4y(1 - y^2) = 0 \Rightarrow y = 0, \pm 1.$$

Hence

$$(0,0), (0,\pm 1), (\pm 1,0), (\pm 1,\pm 1)$$

are the critical points. We find that

$$r = f_{xx} = 4 - 12x^2, s = f_{xy} = 0, t = f_{yy} = -4 + 12y^2.$$

$$rt - s^2 = -16(1 - 3x^2)(1 - 3y^2).$$

Cont...

At the points $(0,1)$ and $(0,-1)$ we have

$$rt - s^2 = 32 > 0, r = 4 > 0,$$

therefore the points $(0,1)$ and $(0,-1)$ are point of relative minima and the minimum value of function is -1 .

At the points $(-1,0)$ and $(1,0)$ we have

$$rt - s^2 = 32 > 0, r = -8 < 0,$$

therefore the points $(0,1)$ and $(0,-1)$ are point of relative maxima and the maximum value of function is 1 .

At the point $(0,0)$ we have $rt - s^2 = -16 < 0$, and at $(\pm 1, \pm 1)$ we have

$$rt - s^2 = -64 < 0,$$

hence the points $(0,0)$ and $(\pm 1, \pm 1)$ are point of neither minima nor maxima.

Practice Problems

Example 1: Find the extreme values of the function

$$f(x,y) = x^3 + y^3 + 3axy, \quad a \in \mathbb{R}$$

Ans: Minimum at (a, a) when $a > 0$ with minimum value $-a^3$ and maximum at (a, a) when $a < 0$. with maximum value a^3 .

Example 2: Examine the extreme points for the maxima and minima of the following functions:

$$f(x,y) = x^3 + y^3 - 63(x+y) + 12xy.$$

Ans: minimum at $(-3, 0)$ with minimum value 3.

Thank
you !